

AI-Driven Wildlife Recognition Under Low-Light Conditions

Data Augmentation and Attention-Enhanced YOLOv8 for Camera-Trap Animal Detection

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1. Introduction & Motivation

Automated monitoring via **infrared (IR)** camera traps is crucial for biodiversity conservation. However, nighttime processing is restricted by severe **visual limitations**:

Degraded Signal: Severe loss of contrast and color.

Habitat Occlusions: Obscuration by branches and shadows.

High Ambiguity: Near-identical IR silhouettes (e.g., Fox vs. Dog).

Data Scarcity: Long-tailed imbalance in rare species.



Figure 1: Representative nighttime IR frames showing low contrast and visual ambiguity.

2. Dataset & Quality Control

Collected from the **Northeast Tiger and Leopard National Park**:

Scale: 10,344 nighttime images across 17 distinct species.

Format Standardisation: Pascal VOC XML to YOLO (.txt).

QC Fix: Clipped 1,233 out-of-bound boxes to [0, W] X [0, H] to eliminate gradient anomalies.

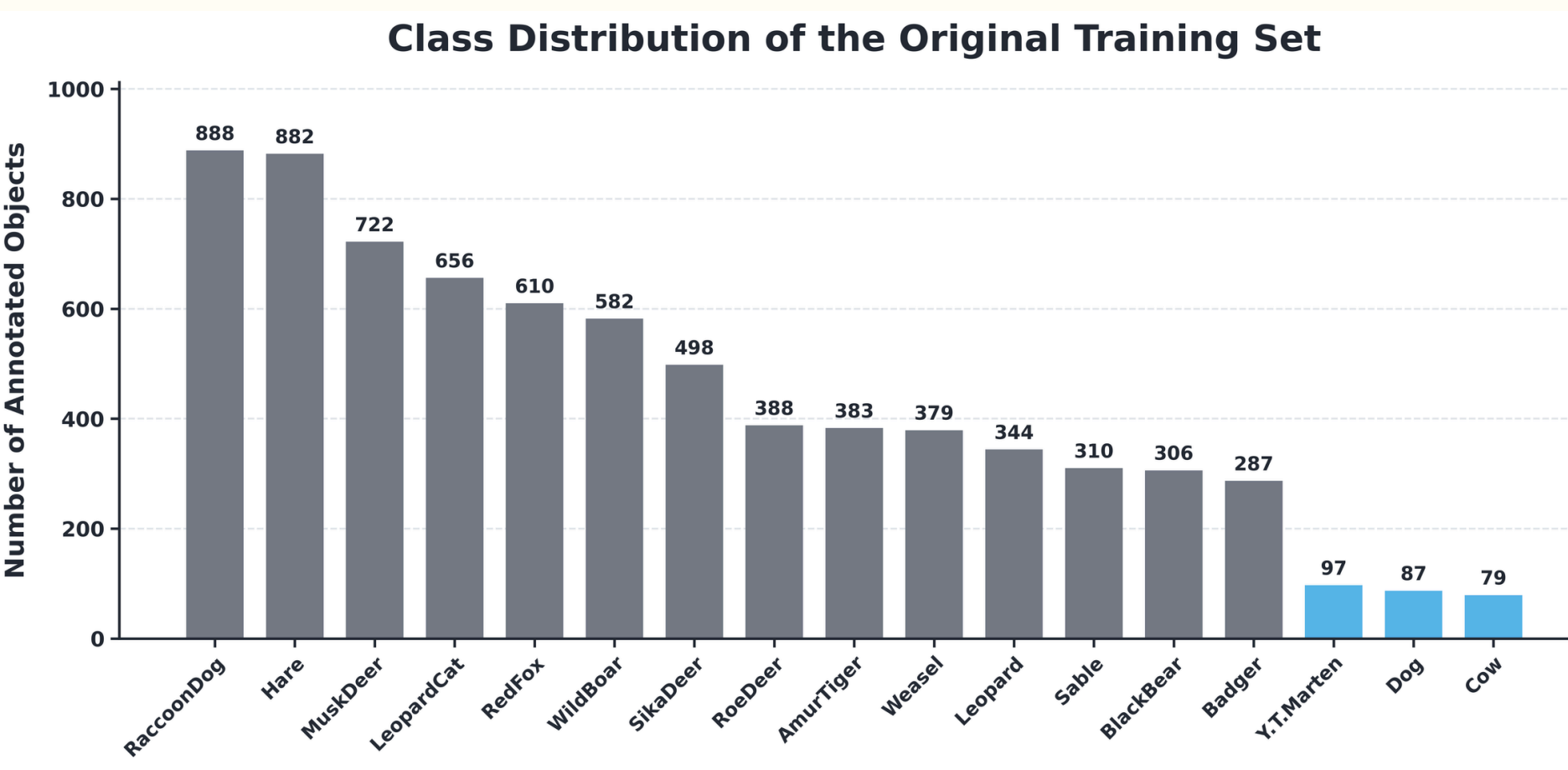


Figure 2: Long-Tails Class Distribution across 17 Species.

3. Two-Stage Targeted Augmentation

Stage 1 (V1): Photometric Augmentation

Simulates illumination variations for minority classes without altering box geometries.

Methods: Dynamic Gamma Correction, CLAHE, and Brightness/Contrast shifts.

Class-Aware Strategy: 80% probability for minority species (e.g., Dog, Cow) vs 30% for majority classes.

Stage 2 (V2): BBox-Aware CutOut

Simulates partial vegetation/branch occlusion within target bounding boxes.

Occlusion Constraint: Mask area $\leq 20\%$ of BBox; filled with local median pixels to avoid edge noise.

Hard-Class Targeting: 60% probability for hard classes (Dog, Roe Deer, Badger, Y.T. Marten) vs 15% for normal samples.

5. Experimental Results

Fixed Validation & Test Sets | 100 Epochs | 640×640 | Seed 42

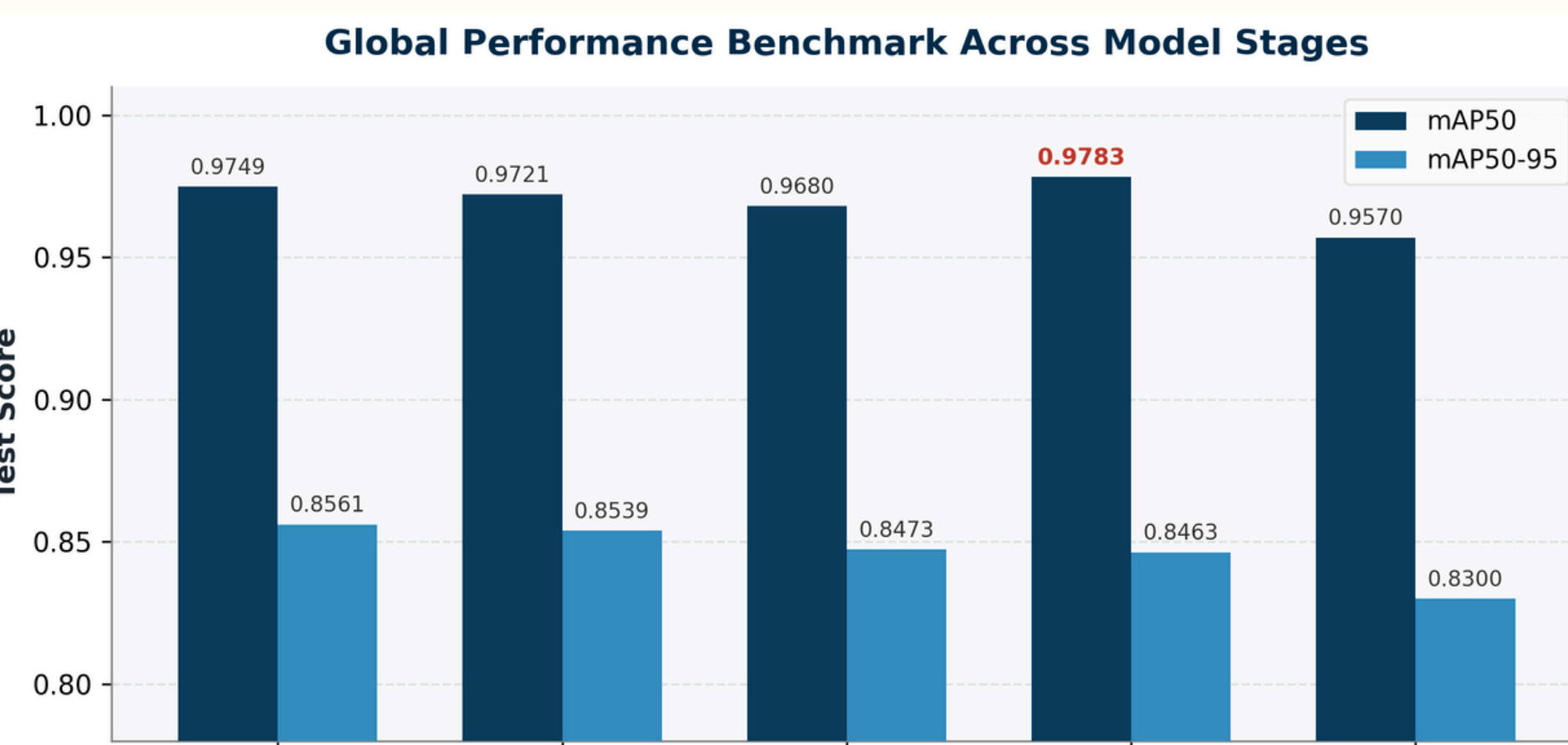


Figure 5a: Global test performance across model stages.

Key result: V3 achieved the highest **mAP50 (0.9783)**, while the baseline retained the highest **mAP50-95 (0.8561)**.

	Difficult-Class Recall Across Model Stages			Δ V2 $\rightarrow$ V3
	Raw YOLOv8s	V2 + Augmentation	V3 + CA-CBAM	
Dog	0.54	0.69	0.92	$\uparrow +23$ pp
Y.T. Marten	0.64	0.64	0.64	$\rightarrow 0$ pp
Cow	0.67	0.92	0.83	$\downarrow -9$ pp
Sable	0.82	0.70	0.84	$\uparrow +14$ pp
Weasel	0.83	0.87	0.93	$\uparrow +6$ pp

Five lowest-recall classes under the Raw YOLOv8s baseline on the frozen test set.

Figure 5b: Difficult-class recall across selected model stages.

Class-specific gains: CA-CBAM improved recall over V2 for **Dog (+23 pp)**, **Sable (+14 pp)**, and **Weasel (+6 pp)**.

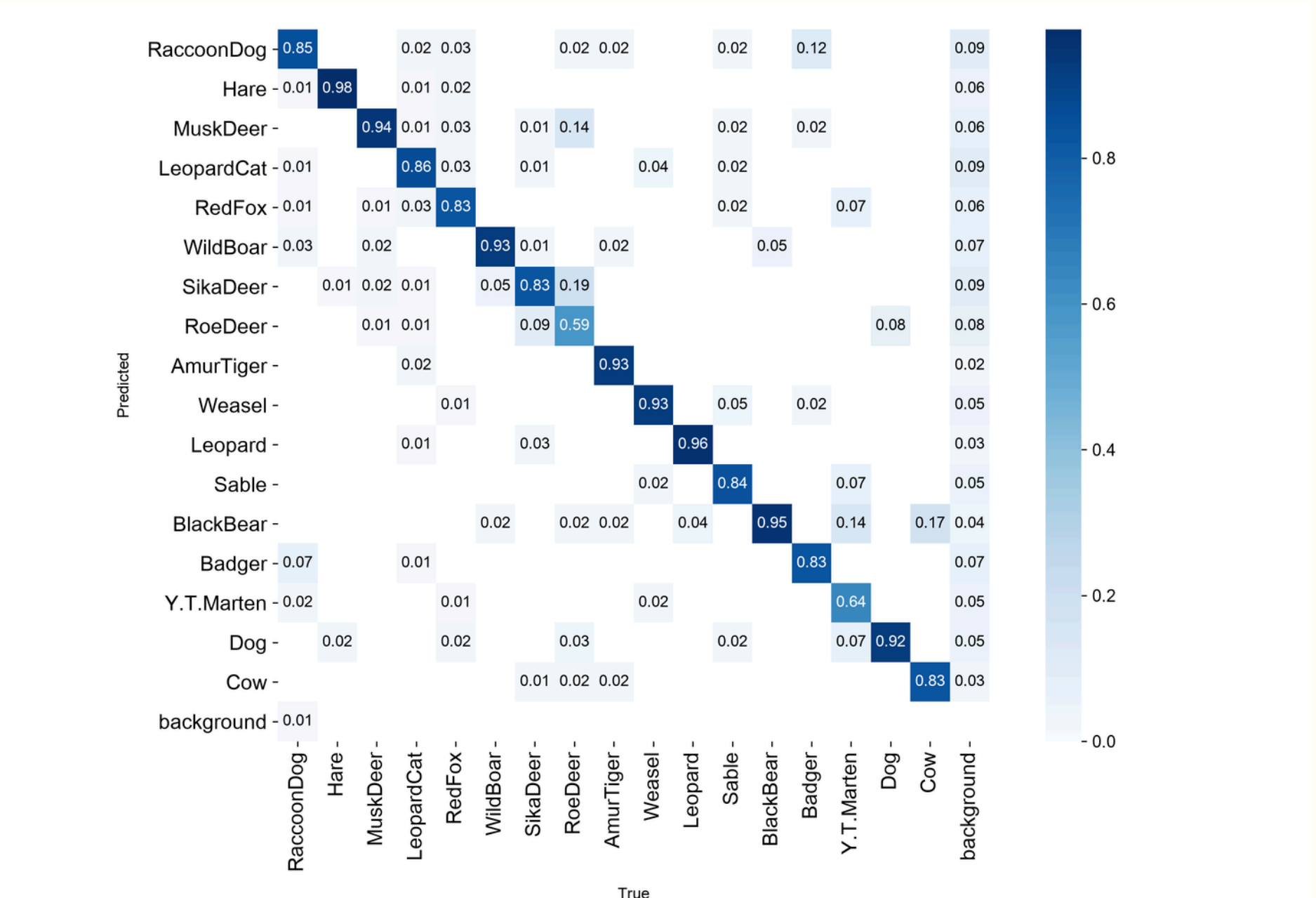
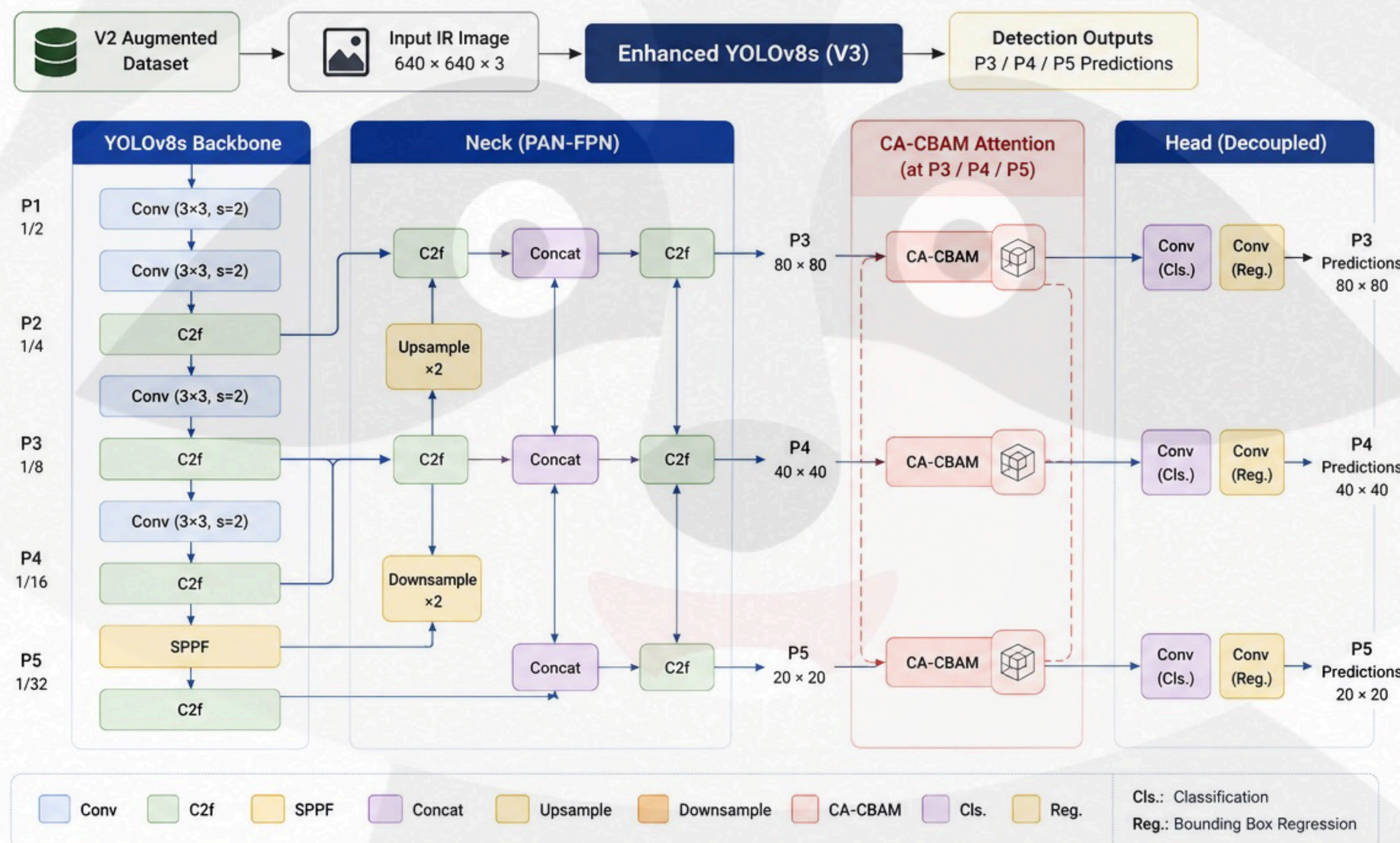


Figure 5c: Normalized test confusion matrix for V3 (CA-CBAM).

Remaining challenge: Y.T. Marten remained difficult to detect (**Recall = 0.64**).

4. Model Architecture & Implementation

Enhanced YOLOv8s Architecture (V3) with CA-CBAM Attention & Focal Loss



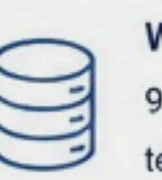
Model Design & Validation



CA-CBAM
Multi-scale attention applied at P3 / P4 / P5 to enhance spatial and channel representation.



Focal Loss
 $\gamma = 2.0, \alpha = 0.25$ for hard-sample learning.



Weight Transfer
97 / 388 $\rightarrow$ 349 / 388 tensors after index-aware remapping.



Validation
Correct pretrained initialization verified before retraining.



Key Takeaway

The proposed architecture enhances YOLOv8s by integrating CA-CBAM attention at multi-scale features (P3/P4/P5) and leveraging Focal Loss to improve detection of nocturnal and rare wildlife under challenging IR conditions.

6. Error Analysis

Selected V3 Test Cases



Figure 6: Representative V3 predictions on the frozen test set.

Misclassification

GT: RedFox
Pred: AmurTiger (0.30)

Correct Detection

GT: RedFox
Pred: RedFox (0.90)

Key Observation: Night time visual ambiguity can still cause inter-class confusion in the final V3 model. The RedFox $\rightarrow$ AmurTiger error likely stems from similar body silhouette and comparable IR brightness at this distance, causing the model to weight coarse shape over fine texture cues that are barely visible at night.

7. Key Insights & Conclusion

- CA-CBAM improved overall detection:** Achieved the highest mAP50 (0.9783), demonstrating strong performance in challenging low-light conditions.
- Strong gains for difficult classes:** Improved recall for Dog (+23 pp), Sable (+14 pp), and Weasel (+6 pp), showing better handling of ambiguous IR images.
- Attention outperformed loss reweighting:** CA-CBAM produced targeted class-level gains, while Focal Loss provided no additional improvement.

Future Work:

- Controlled ablation & hard-case learning:** Separately evaluate CA-CBAM and Focal Loss, alongside class-specific oversampling and hard-negative mining.
- Improve localization & robustness:** Investigate the mAP50-mAP50-95 gap and explore methods to improve bounding-box precision and generalization to challenging IR scenes.